PAPER_163 — UQFF Modular Architecture: Decomposed Compressed MUGE Functions

Session: *47 |* ***Date:*** *March 13, 2026 |* ***Thread:*** *7f9068 |* ***Domain:*** *§2.3*

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Abstract

This paper establishes the **modular decomposition** of the 9-term Compressed MUGE into individual callable functions, each representing a single physical correction term. This transforms the monolithic `compute_compressed_MUGE()` function (SOURCE4) into an auditable, unit-testable architecture where each correction can be independently validated against observational data. Eight decomposed functions are cataloged with their governing equations and parameter dependencies.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. Original 9-Term Compressed MUGE (PAPER_090)

```
g_{comp}(r,t) = g_0 \cdot (1 + H_0 t) \cdot (1 - B/B_{crit}) \cdot f_{env}(r)
```

```
+ \Lambda c^2/3 + \frac{\hbar}{\Delta x \Delta p} \cdot \int\psi \cdot \frac{2\pi}{t_H}
```

```
+ \rho_{fluid} V g_{loc} + (M + M_{DM}) \cdot \left(\frac{\delta\rho}{\rho} +
\frac{3GM}{r^3}\right)
```

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2. Decomposed Function Architecture

Function 1: Base Newtonian Gravity

```
g_{base}(M, r) = \frac{G \cdot M}{r^2}
```

```
double compute_compressed_base(const MUGESystem& sys) {
return G * sys.M / (sys.r * sys.r);
}
```

Function 2: Hubble Expansion Correction

```
g_{exp}(t) = 1 + H_0 \cdot t
```

```
H_0 = 67.4\, \text{km/s/Mpc} = 2.185 \times 10^{-18}\, \text{s}^{-1}
```

```
double compute_compressed_expansion(const MUGESystem& sys) {
const double H0 = 2.185e-18; // s^-1 (Planck 2018)
```

```
return 1.0 + H0 * sys.t;
}
```

Function 3: Magnetic Suppression Adjustment

```
f_{super} = 1 - B/B_{crit}
```

```
double compute_compressed_super_adj(const MUGESystem& sys) {
return 1.0 - sys.B / sys.B_crit;
}
```

Function 4: Envelope Confinement

```
f_{env}(r) = \begin{cases} 1 & r \leq R_{env} \\ e^{-(r-R_{env})/R_{env}} & r > R_{env}
\end{cases}
```

```
double compute_compressed_envelope(const MUGESystem& sys) {
if (sys.r <= sys.R_env)
return 1.0;
return std::exp(-(sys.r - sys.R_env) / sys.R_env);
}
```

Function 5: Cosmological Constant Term

```
g_{cosm} = \frac{\Lambda c^2}{3}
```

```
\Lambda = 1.1 \times 10^{-52}\, \text{m}^{-2}
```

```
double compute_compressed_cosm(double Lambda = 1.1e-52) {
const double c = 2.998e8;
return Lambda * c * c / 3.0;
}
```

Function 6: Quantum Uncertainty Term

```
g_{quantum} = \frac{\hbar}{\Delta x \Delta p} \cdot \int\psi \cdot \frac{2\pi}{t_H}
```

where `t_H = 1/H_0` is the Hubble time and `\int\psi` is the integrated wave function amplitude.

```
double compute_compressed_quantum(double Delta_x, double Delta_p, double psi_int) {
const double hbar = 1.055e-34;
const double H0 = 2.185e-18;
const double t_H = 1.0 / H0;
return (hbar / (Delta_x * Delta_p)) * psi_int * (2.0 * M_PI / t_H);
}
```

Function 7: Buoyant Fluid Term

```
g_{fluid} = \rho_{fluid} \cdot V_{body} \cdot g_{local}
```

```
double compute_compressed_fluid(const MUGESystem& sys) {
return sys.rho_fluid * sys.V_body * sys.g_local;
}
```

Function 8: Dark Matter Perturbation

```
g_{pert} = (M + M_{DM}) \cdot \left(\frac{\delta\rho}{\rho} + \frac{3GM}{r^3}\right)
```

```
double compute_compressed_perturbation(const MUGESystem& sys) {
double tidal = 3.0 * G * sys.M / (sys.r * sys.r * sys.r);
return (sys.M + sys.M_DM) * (sys.delta_rho_over_rho + tidal);
}
```

---

3. Full Decomposed Master Function

```
double compute_compressed_MUGE_modular(const MUGESystem& sys,
double Delta_x = 1e-10,
double Delta_p = 1e-24,
double psi_int = 1.0) {
double g_base = compute_compressed_base(sys);
double g_exp = compute_compressed_expansion(sys);
double g_super = compute_compressed_super_adj(sys);
double g_env = compute_compressed_envelope(sys);
double g_cosm = compute_compressed_cosm();
double g_quant = compute_compressed_quantum(Delta_x, Delta_p, psi_int);
double g_fluid = compute_compressed_fluid(sys);
double g_pert = compute_compressed_perturbation(sys);
return g_base * g_exp * g_super * g_env + g_cosm + g_quant + g_fluid + g_pert;
}
```

---

4. Unit Test Matrix

Function	Test Input	Expected Output	Tolerance
`compute_base`	M=1e30, r=1e11	6.67×10■	1%
`compute_expansion`	t=0	1.0000	<0.01%
`compute_super_adj`	B=0	1.0000	exact
`compute_super_adj`	B=B_crit	0.0000	exact
`compute_cosm`	Λ=1.1e-52	3.293×10^{-3}■	1%
`compute_quantum`	ΔxΔp=■/2, ψ=1	varies	order of mag.
`compute_fluid`	ρ=1.29, V=1, g=9.81	12.66	1%
`compute_perturbation`	δρ/ρ=0.1, M_DM/M=10	varies	order of mag.

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5. CP Integration

CP1 (`CondensedPhysics.py`**):** Add `CompressedMUGEModularCalculator` with 8 sub-methods. **CP2 (**`CondensedPhysics2.py`**):** Refactor existing `compute_compressed_MUGE` into modular form.

Status: ■ Complete | **CP Stage:** CP1/CP2 **Supersedes:** N/A (extends PAPER*090)* | ***Related:*** *PAPER*090 (9-term compressed), PAPER*158 (hybrid blending uses g*comp), PAPER_164 (quantum term calibration from CERN)
